Sethuraman K

AI / ML Engineer | Chennai, India | Phone: 8667562719 | Email: sethuramanvr046@gmail.com

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PROFESSIONAL SUMMARY

AI / ML Engineer with experience building, deploying, and maintaining intelligent systems across computer vision, backend services, cloud platforms, and automation workflows. Hands-on experience working in startup environments on ML-powered features, cloud-deployed services, and AI-enabled applications. Strong exposure to Python, AWS, GCP, Docker and LLM integration. Able to adapt quickly, solve production issues, and contribute to real-world AI systems.

WORK EXPERIENCE

AI / ML Engineer | Amalaan (BugFlows) – Chennai, India | June 2024 – Dec 2025

- Developed AI-driven automation features for issue tracking platforms including Jira, YouTrack, and Bugzilla.
- Built backend services and ML pipelines using Python and Node.js.
- Integrated LLM-based systems for automation, classification, and intelligence workflows.
- Deployed and maintained applications on AWS and GCP.
- Worked on CI/CD pipelines and containerization using Docker.
- Contributed to frontend features using React.
- Collaborated with cross-functional teams to design and implement production systems.

Junior AI Specialist | Abelyf – Chennai, India | Nov 2023 – June 2024

- Developed multi-camera tracking and object detection systems using computer vision models.
- Built real-time backend logic for synchronization and monitoring systems.
- Implemented automation workflows using Python for robotics-related systems.
- Worked on smoke detection and healthcare-related applications.
- Integrated ML models onto hardware devices.

Intern – Computer Vision | IOTIOT.in – Chennai, India | Oct 2022 – Dec 2022

- Developed object detection and image processing solutions using OpenCV.
- Assisted in automation and computer vision research activities.
- Contributed to project testing and deployment tasks.

EDUCATION

Bachelor of Engineering in Biomedical Engineering
Sathyabama University, Chennai | 2020 – 2024 | CGPA: 8.0

PROJECTS

AI Interviewer with Custom Avatar & Malpractice Detection (Github)

- Built a full-stack AI interviewer system using React and Node.js.
- Integrated LLM-based interview automation and dynamic question handling.
- Implemented malpractice detection using computer vision (multiple persons, mobile phone usage).

- Added voice cloning and lip synchronization for realistic avatar interaction.
- Managed backend services using Python and REST APIs.

Wheelchair Automation System (Github)

- Designed an autonomous wheelchair navigation system using computer vision.
- Implemented lane detection and curvature estimation for movement control.
- Integrated gaze tracking and hand-gesture control for accessibility.
- Developed image processing pipelines for real-time decision making.

Autism Student Monitoring System (Github)

- Developed a real-time behavior and emotion monitoring platform for students with autism.
- Implemented object detection and behavior classification using YOLO and CNN models.
- Integrated facial emotion recognition and face identification features.
- Built data logging and visualization using CSV storage and charts.
- Created both web interface (Streamlit) and desktop application (Tkinter).
- Containerized application using Docker for easy deployment.

AI-Powered Avatar Assistant (Lip-Sync Chatbot)

- Built an interactive AI avatar assistant using deep-learning-based lip-sync.
- Integrated text-to-speech and audio-driven facial animation using Wav2Lip.
- Developed chatbot logic and UI for real-time conversation.
- Created an end-to-end pipeline for speech generation and avatar animation.

3D Fighting Game with Custom Avatar (Game)

- Built a 3D fighting game using a custom-created avatar pipeline.
- Generated personalized 3D avatars from images using AI tools and Blender.
- Integrated auto-rigging and animation using Mixamo.
- Implemented character movement and combat mechanics in Unity.
- Designed animation controllers and game logic for interactive gameplay.

SKILLS

Programming: Python | ML: PyTorch, TensorFlow, CNNs, Computer Vision, GenAI
Tools: OpenCV, YOLO, Unity (Basic), Unreal Engine (Basic), Blender | Cloud: AWS, GCP | DevOps: Docker, CI/CD, GitHub Actions | Frontend: React, Html | Automation: Selenium, Py Tesseract

ACHIEVEMENTS

- Published author of the book “Hands-On Computer Vision – Volume 1”.
- Creator of the cvworld Python package.
- Udemy instructor delivering computer vision content.